WTW158 CALCULUS
SEMESTER TEST 1 / SEMESTERTOETS 1

Date / Datum: March /Maart 2017 Marks / Punte : 45 Time / Tyd : 2 hours / uur

MARKS FOR SECTIONS B & C PUNTE VIR AFDELINGS B & C	MEMO
	27

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SURNAME / VAN | | | | | | | | | | | | | | | |

FIRST NAMES / VOORNAME ____________________

STUDENT NUMBER
STUDENTENOMMER | | | | | | | | |

SIGNATURE
HANDTEKENING ____________________

CONTACT NUMBER
KONTAKNOMMER | | | | | | | | | | | |

CIRCLE YOUR GROUP NUMBER / OMKRING JOU LESINGGROEPNOMMER.

1	2	3	4	5
Ms Dreyer	Prof Harding	Dr Garba	Dr van der Hoff	Dr Ohlhoff / Dr Jooste

READ THE FOLLOWING INSTRUCTIONS

1. The paper consists of pages 1 to 10. Check whether your paper is complete.
2. **No calculator may be used.**
3. No question paper may be removed from the venue.

LEES DIE VOLGENDE INSTRUKSIES

1. Die vraestel bestaan uit bladsye 1 tot 10. Kontroleer of u vraestel volledig is.
2. **Geen sakrekenaar mag gebruik word nie.**
3. Geen vraestel mag uit die lokaal geneem word nie.

Section A: / Afdeling A:

Marks: 18 / Punte: 18 (1.5 per question / vraag)

1. The questions in Section A must be completed on **SIDE 2** of the optical reader form in SOFT PENCIL.
2. First circle your answers on this paper and then transfer these to the optical reader form.
3. There is only ONE correct answer per question.

1. Die vrae in Afdeling A moet op **KANT 2** van die merkleesvorm met 'n SAGTE POTLOOD voltooi word.
2. Omkring eers jou antwoorde op hierdie vraestel en dra dit dan oor na die merkleesvorm.
3. Daar is slegs EEN korrekte antwoord per vraag.

Question 1 / Vraag 1

Consider / Beskou

$$f(x) = \frac{|x|}{\sqrt{2+x-x^2}}$$

The domain of the function is / Die definisieversameling van die funksie is

a)	$(-1,2)$	b)	$(-\infty,-1)\cup(2,\infty)$
c)	$(-2,1)$	d)	$(-\infty,-2)\cup(1,\infty)$
e)	None of these / Geen van hierdie		

Question 2 / Vraag 2

If / As $2\ln x - \ln(x+2) = 0$, then / dan is

(a)	$x=-1$ or/of $x=2$	(b)	$x=0$	(c)	$x=1\pm\sqrt{2}$
(d)	$x=-1+\sqrt{2}$	(e)	$x=2$	(f)	None of these / Geen van hierdie

Question 3 / Vraag 3

cosec $\left(\frac{39\pi}{2}\right) =$

(a)	$\frac{\sqrt{3}}{2}$	(b)	-1	(c)	$-\sqrt{2}$
(d)	1	(e)	0	(f)	None of these / Geen van hierdie

Question 4 / Vraag 4

$\cos^{-1}(-\frac{1}{2}) =$ (Notation: $\cos^{-1}$ = arccos / Notasie: $\cos^{-1}$ = bgcos)

(a)	$-\frac{\pi}{6}$	(b)	$-\frac{\pi}{3}$
(c)	$\frac{2\pi}{3}$	(d)	$\frac{5\pi}{6}$
(e)	None of these / Geen van hierdie		

Question 5 / Vraag 5

Given / Gegee $y = e^{-x}$.
The equation of the graph that results from shifting 2 units right is
Die vergelyking van die grafiek wat verkry word deur 2 eenhede na regs te skuif is

(a)	$e^{-x} - 2$	(b)	e^{-x+2}
(c)	e^{-x-2}	(d)	$2e^{-x-2}$
(e)	None of these / Geen van hierdie		

Question 6 / Vraag 6

If / Indien $\sin\theta = -\frac{1}{2}$ and /en $0 \leq \theta < 2\pi$, then / dan is $\theta =$

(a)	$\frac{\pi}{3}$ or / of $\frac{2\pi}{3}$	(b)	$-\frac{\pi}{6}$ or / of $\frac{5\pi}{6}$
(c)	$\frac{7\pi}{6} + 2k\pi$ or / of $\frac{11\pi}{6} + 2k\pi, k \in \mathbb{Z}$	(d)	$\frac{7\pi}{6}$ or / of $\frac{11\pi}{6}$
(e)	None of these / Geen van hierdie		

Question 7 / Vraag 7

If $f : A \to B$ and f^{-1} is the inverse of f then $(f \circ f^{-1})(x) = x$ for
As $f : A \to B$ en f^{-1} is die inverse van f dan is $(f \circ f^{-1})(x) = x$ vir

(a)	$x \in \mathbb{R}$	(b)	$x \in A$
(c)	$x \in B$	(d)	$x = 1$
(e)	None of these / Geen van hierdie		

Question 8 / Vraag 8

The solution of / Die oplossing van $\frac{\ln(x-2)}{x^2+1} \leq 0$ is

(a)	$x \in (2,3]$	(b)	$x \in (-\infty, 2]$	(c)	$x \in (0,2]$
(d)	$x \in (-\infty, 3]$	(e)	$x \in (-\infty, 3)$	(f)	None of these / Geen van hierdie

Question 9 / Vraag 9

The solution of / Die oplossing van $\sin x + \cos x = 0$ for/vir $x \in [0, 2\pi)$ is

(a)	$x = 0$	(b)	$x = \frac{\pi}{2}$	(c)	$x = \frac{3\pi}{4}$ or/of $x = \frac{7\pi}{4}$
(d)	$x = \frac{\pi}{4}$ or/of $x = -\frac{3\pi}{4}$	(e)	$x = \frac{\pi}{4}$ or/of $x = \frac{5\pi}{4}$	(f)	None of these / Geen van hierdie

Question 10 / Vraag 10

Which one of the following is true for the function / Watter een van die volgende is waar vir die funksie:
$f(x) = \frac{x-2}{x^2+x-6}$?

(a) f has a vertical asymptote $x = 2$ / f het 'n vertikale asymptoot $x = 2$
(b) f has a vertical asymptote $x = -3$ / f het 'n vertikale asymptoot $x = -3$
(c) f has vertical asymptotes $x = 2$ and $x = -3$ / f het 'n vertikale asymptoot $x = 2$ en $x = -3$
(d) f has no vertical asymptotes / f het geen vertikale asymptote nie.
(e) None of these / Geen van hierdie

Question 11 / Vraag 11

The solution of / Die oplossing van $\ln(\ln x) = 1$ is

(a) $x = 0$	(b) $x = 1$	(c) $x = e^e$	(d) $x = e$
(e) None of these / Geen van hierdie			

Question 12 / Vraag 12

The average rate of change of $f(x) = x^2 + 3$ over the interval $[1,3]$ is
Die gemiddelde tempo van verandering van $f(x) = x^2 + 3$ oor die interval $[1,3]$ is

(a) 2	(b) 4	(c) 6	(d) 8
(e) None of the above / Geen van hierdie			

Section B Marks: 12 / Afdeling B Punte: 12

1. This section must be completed IN PEN in this paper!
2. Fill in the answer only in the space provided. Rough work can be done on the facing page and will not be marked.

1. Hierdie afdeling moet MET PEN op hierdie vraestel gedoen word.
2. Vul slegs die antwoord in die spasie in. Rofwerk kan op die teenblad gedoen word en sal nie gemerk word nie.

Question 1 / Vraag 1

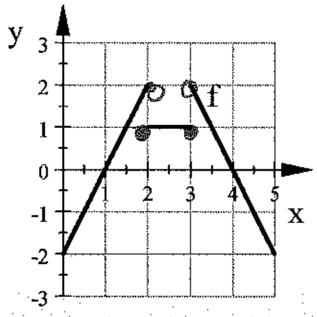
Write the function $f(x) = |1 - 4x^2|$ as a piece-wise defined function.
Skryf die funksie $f(x) = |1 - 4x^2|$ as 'n stuksgewys gedefinieerde funksie.

$$f(x) = \begin{cases} 1 - 4x^2 & \text{if / as } x \in \left[-\frac{1}{2}, \frac{1}{2}\right] \\ -(1-4x^2) & \text{if / as } x \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right) \end{cases}$$

2

Question 2 / Vraag 2

2.1 Let / Laat $y = f(x)$



Write $y = f(x)$ as a piecewise defined function. Use interval notation.
Skryf $y = f(x)$ as 'n stuksgewys gedefineerde funksie.Gebruik intervalnotasie.

$$f(x) = \begin{cases} 2x - 2 & \text{if} \quad x \in [0, 2) \\ 1 & \text{if} \quad x \in [2, 3] \\ -2x + 8 & \text{if} \quad x \in (3, 5] \end{cases}$$

2

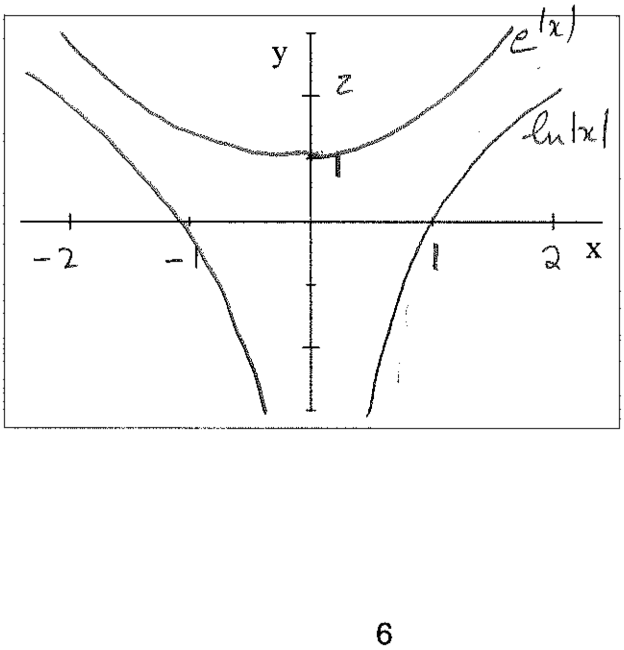
2.2 Find / Bepaal $f \circ f(3)$

$f \circ f(3) = 0$. or 0

1

Question 3 / Vraag 3

Sketch the graphs of $y_1 = \ln|x|$ and $y_2 = e^{|x|}$ on the same set of axes. Clearly name the functions and indicate all intercepts with the axes.
Skets die grafieke van $y_1 = \ln|x|$ and $y_2 = e^{|x|}$ op dieselfde assestelsel. Benoem die funksies duidelik en dui alle snypunte met die asse aan.



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Question 4 / Vraag 4 [1]

Write as one expression: / Skryf as een uitdrukking:

$\frac{1}{3}\ln(x+2)^3 + \frac{1}{2}\left[\ln x - \ln(x^2+3x+2)^2\right] =$

$\ln\left(\frac{\sqrt{x}}{x+1}\right)$

1

Question 5 / Vraag 5

A population grows according to the formula $P(t) = 10e^{0.05t}$ where $P(t)$ is measured in millions and t is measured in years. When will the population exceed the 20 million mark? (Assume that $\ln 2 \simeq 0.69$)

'n Bevolking groei volgens die formule $P(t) = 10e^{0.05t}$ waar $P(t)$ gemeet word in miljoene en t gemeet word in jare. Wanneer sal die bevolking die 20 miljoen merk oorskrei? (Gestel dat $\ln 2 \simeq 0.69$)

13,8 years / $13\frac{4}{5}$ years

1

Question 6 / Vraag 6

Given / Gegee: $g(x) = \sin^{-1}(2x+1)$ then / dan is

$D_g = [-1, 0]$ $W_g = [-\pi/2, \pi/2]$

2

Question 7 / Vraag 7

$\lim_{x \to 4^+} \frac{1}{4-x} =$

$-\infty$

1

Afdeling C Marks: 15 / Section C Punte: 15

1. The question in Section C must be completed on this paper.
2. All calculations must be shown and will be marked.
3. For more than the available space, use the facing page and clearly indicate it.
4. Attention should be paid to writing in a mathematically sound way.

1. Die vrae in Afdeling C moet op die vraestel voltooi word.
2. Alle berekeninge moet getoon word en sal gemerk word.
3. Vir meer as die beskikbare ruimte, gebruik die teenblad en dui dit duidelik aan.
4. Maak seker dat jou skryfwyse wiskundig korrek is.

Question 1 / Vraag 1

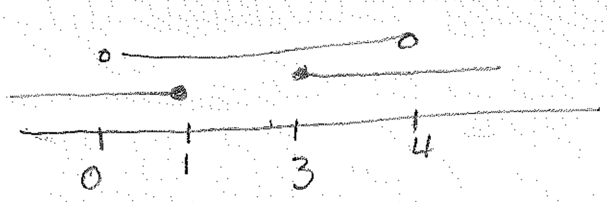
Solve the inequality $1 \le |2-x| < \cancel{8}\ 2$. Show all calculations.
Los die ongelykheid $1 \le |2-x| < \cancel{8}\ 2$ op. Toon alle berekeninge.

$|2-x| \geq 1$ and $|2-x| < 2$

$\Rightarrow 2-x \geq 1$ or $2-x \leq -1$ and $-2 < 2-x < 2$

$\Rightarrow x \leq 1$ or $x \geq 3$ and $-4 < -x < 0$

$0 < x < 4$



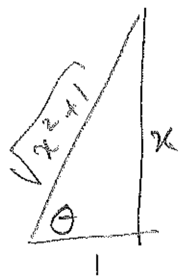
$\Rightarrow x \in (0, 1] \cup [3, 4)$

3

Question 2 / Vraag 2

Determine / Bepaal: $\sin(2\tan^{-1}x)$ θ

$= 2\sin(\tan^{-1}x) \cdot \cos(\tan^{-1}x)$



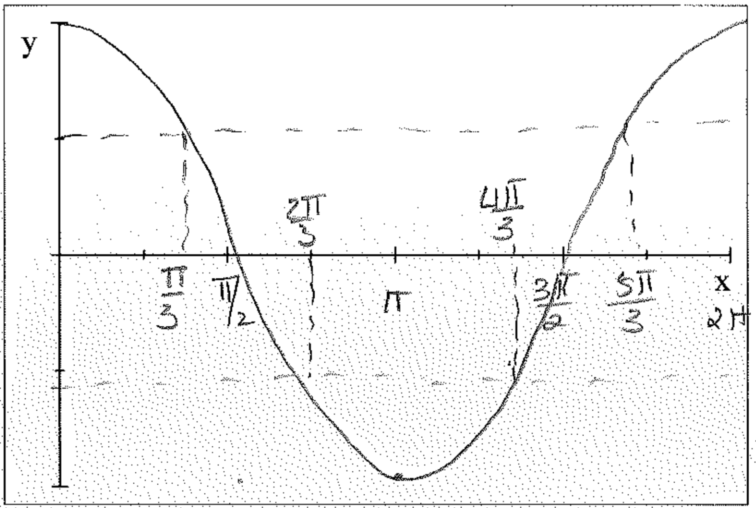
$= 2 \cdot \dfrac{x}{\sqrt{x^2+1}} \cdot \dfrac{1}{\sqrt{x^2+1}}$

$= \dfrac{2x}{x^2+1}$

2

Question 3 / Vraag 3

Use a sketch to solve the inequality $|\cos x| \geq \frac{1}{2}$ for $x \in [0, 2\pi]$.
Gebruik 'n skets om die ongelykheid $|\cos x| \geq \frac{1}{2}$ op te los vir $x \in [0, 2\pi]$.



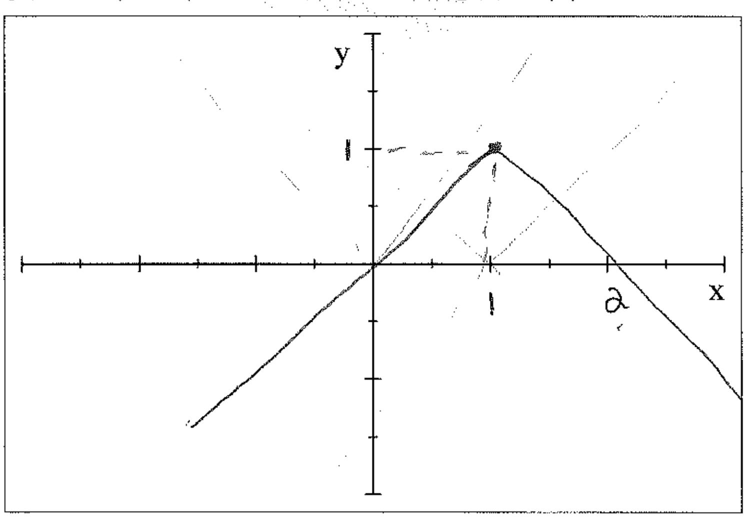
Solution: / Oplossing:

$x \in \left[0, \frac{\pi}{3}\right] \cup \left[\frac{2\pi}{3}, \frac{4\pi}{3}\right] \cup \left[\frac{5\pi}{3}, 2\pi\right]$

3

Question 4 / Vraag 4

Sketch the function $f(x) = -|x-1| + 1$ through transforming the function $g(x) = |x|$.
Skets die funksie $f(x) = -|x-1| + 1$ deur die funkie $g(x) = |x|$ te transformeer.



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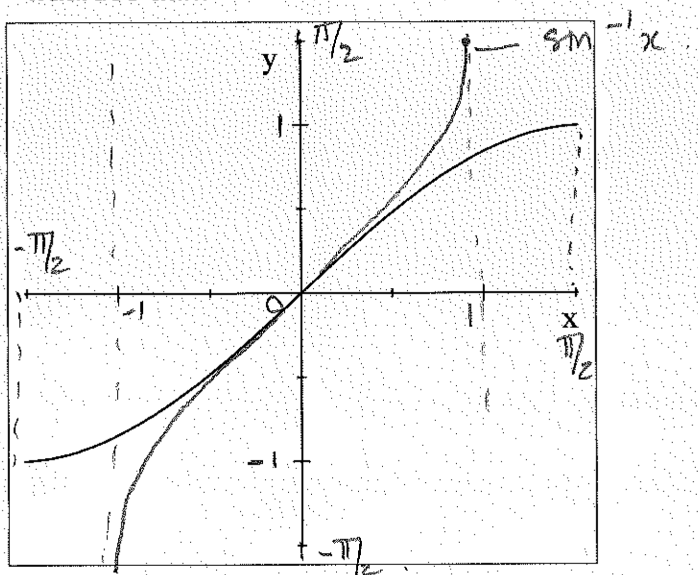
Question 5 / Vraag 5

5.1 Complete the definition / Voltooi die definisie:

$y = \arcsin x \Leftrightarrow$ ......$\sin y = x$......where / waar...$y \in [-\pi/2, \pi/2]$......

1

5.2 The function $f(x) = \sin x$ is given on an interval on which it is one-to-one. Sketch the function $g(x) = \arcsin x = \sin^{-1}x$ on the same set of axes and on the applicable interval. Show all significant x – and y – values.
Die funksie $f(x) = \sin x$ word gegee op 'n interval waarop dit een-tot-een is. Skets die funksie $g(x) = \arcsin x = \sin^{-1}x$ op dieselfde assestelsel en op die betrokke interval. Dui alle relevante x – en y – waardes aan.



2

Question 6 / Vraag 6

If f is an even function and g is an odd function, show that $f \circ g$ is an even function.
As f 'n ewe funksie is en g 'n onewe funksie, toon aan dat $f \circ g$ 'n ewe funksie is.

$$f \circ g(-x) = f(g(-x))$$
$$= f(-g(x)) \quad g \text{ odd}$$
$$= f(g(x)) \quad f \text{ even}$$

2